SE311 SPRING 2025-2026 / 25-02-2026 && 26-02-2026
Lab Work 01 – SOLID
Goal: Practice on SOLID Principles

Download the **SmartHomeTest.java** file from Blackboard.

Scan and analyze the underlying structure of the existing code. There are some issues related with the **SOLID** Principles in the code.

Identify the violating parts and fix them according to **SOLID** principles**.**

HINTS:

Is it really SmartLight's concern to record video? What happens if you call recordVideo() on a light bulb?

What will happen when **`CentralHub`** wants to switch from using **`SmartLight`** to a **`SecurityCamera`** or, in the future, manage all of the devices together?

Implement another device into the system called **`SmartThermostat`** and integrate it to work with the previous devices in **`CentralHub`**.

Does **`SmartLight`** record video?

What happens to different coefficients of every single device in `calculateEnergyCost()` method's if blocks in **`CentralHub`** class?